

**LAPORAN PRAKTIKUM
TEKNOLOGI IOT II**

TUGAS 4.7 DAN 5.7



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**TEKNOLOGI REKAYASA INTERNET
JURUSAN TEKNOLOGI INFORMASI
POLITEKNIK NEGERI LAMPUNG**

2026/2027

Link github: <https://github.com/se-karr/praktikum-iot-4-dan-5.git>

Modul 4: Komunikasi Asinkron dan WebSockets

4.7 Tugas: Sistem Kendali Intensitas Cahaya (PWM Slider) via *WebSockets*

A. Alat dan Bahan

Alat dan bahan yang digunakan dalam praktikum ini adalah:

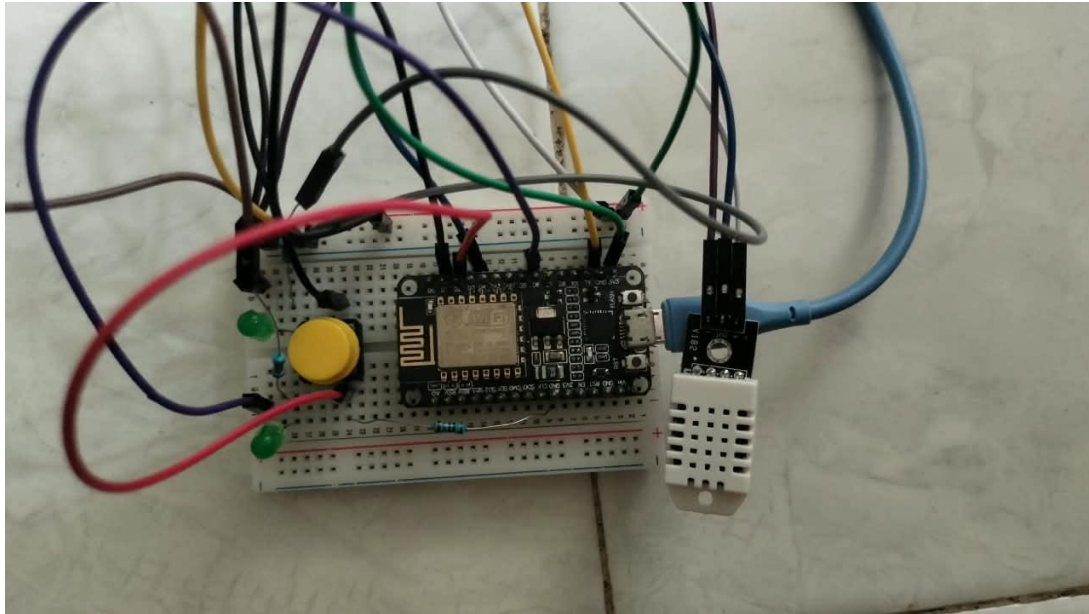
No.	Alat dan Bahan	Jumlah
1	NodeMCU ESP8266	1 buah
2	Sensor DHT11	1 buah
3	LED 5mm	2 buah
4	Resistor 220 Ohm	2 buah
5	Resistor 10k Ohm	1 buah
6	Push Button	1 buah
7	Breadboard	1 buah
8	Kabel jumper	Secukupnya
9	Kabel Micro-USB	1 buah
10	Laptop dengan Arduino IDE	1 unit
11	Jaringan Wi-Fi / Hotspot	1 jaringan

B. Langkah Kerja

1. Perancangan Rangkaian

Langkah pertama adalah menyiapkan NodeMCU ESP8266 dan seluruh komponen yang diperlukan. Sensor DHT11 dihubungkan ke pin D4, LED pertama dihubungkan ke pin D6 melalui resistor 220 Ohm, dan push button dihubungkan ke pin D2 menggunakan rangkaian resistor pull-down 10k Ohm.

Selanjutnya, LED kedua dipasang pada pin D1 melalui resistor 220 Ohm. LED kedua digunakan sebagai aktuator untuk pengujian kendali intensitas cahaya menggunakan PWM.



2. Kode yang digunakan

Berikut disajikan beberapa potongan kode program yang menjadi bagian utama dalam pelaksanaan praktikum. Mengingat jumlah baris kode yang cukup banyak, hanya bagian-bagian penting yang ditampilkan dalam laporan ini. Adapun kode program secara keseluruhan telah diunggah ke repositori GitHub pribadi dan dapat diakses melalui tautan yang telah dicantumkan.

```
4.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...)

4.7.ino
1 #include <ESP8266WiFi.h>
2 #include <ESPAsyncTCP.h>
3 #include <ESPAsyncWebServer.h>
4 #include <DHT.h>
5
6 // Konfigurasi WiFi
7 const char* ssid = "kar";
8 const char* password = "12345678";
9
10 // Konfigurasi Pin
11 const byte dhtPin = 2; // D4
12 const byte buttonPin = 4; // D2
13 const byte ledPin = 12; // D6
14 const byte pwmPin = 5; // D1
15
16 DHT dht(dhtPin, DHT22);
17
18 // Variabel status
19 bool ledState = false;
20 String currentTemp = "--";
21 int nilaiPWM = 0;
22
23 int buttonState = LOW;
24 int lastButtonState = LOW;
25
26 unsigned long lastDebounceTime = 0;
27 const unsigned long debounceDelay = 50;
28 unsigned long lastTime = 0;
29
```

```
4.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...)

4.7.ino
30 // Web Server dan WebSocket
31 AsyncWebServer server(80);
32 AsyncWebSocket ws("/ws");
33
34 // =====
35 // HTML, CSS, DAN JAVASCRIPT
36 // =====
37
38 const char index_html[] PROGMEM = R"rawliteral(
39 <!DOCTYPE html>
40 <html>
41 <head>
42   <meta name="viewport" content="width=device-width, initial-scale=1">
43   <title>Smart Lighting - PWM Slider</title>
44
45   <style>
46     body {
47       font-family: Arial;
48       text-align: center;
49       background-color: #f0f0f0;
50     }
51
52     .card {
53       background-color: white;
54       margin: 20px auto;
55       padding: 20px;
56       max-width: 350px;
57       border-radius: 10px;
58     }
59
```

```
4.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...)

4.7.ino
59
60     button {
61       padding: 15px 30px;
62       font-size: 18px;
63       border: none;
64       border-radius: 5px;
65       cursor: pointer;
66       color: white;
67     }
68
69     .btn-on {
70       background-color: #4CAF50;
71     }
72
73     .btn-off {
74       background-color: #f44336;
75     }
76
77     input[type="range"] {
78       width: 90%;
79     }
80   </style>
81 </head>
82
83 <body>
84
85   <h1>Smart Lighting</h1>
86   <h3>NodeMCU ESP8266 - WebSockets</h3>
87
```

```
4.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12E M...)

4.7.ino
338   lastDebounceTime = millis();
339 }
340
341 if ((millis() - lastDebounceTime) > debounceDelay) {
342   if (reading != buttonState) {
343     buttonState = reading;
344
345     if (buttonState == HIGH) {
346       ledState = !ledState;
347       notifyClients();
348     }
349   }
350 }
351
352 lastButtonState = reading;
353
354 // Membaca suhu setiap 3 detik
355 if ((millis() - lastTime) > 3000) {
356   float t = dht.readTemperature();
357
358   if (!isnan(t)) {
359     currentTemp = String(t);
360     notifyClients();
361   }
362
363   lastTime = millis();
364 }
365 }
366
```

3. Pengujian Sistem

Setelah program diunggah ke NodeMCU, Serial Monitor dibuka dengan baud rate 115200 untuk mengetahui alamat IP perangkat. Alamat IP tersebut kemudian diakses melalui browser yang terhubung ke jaringan Wi-Fi yang sama.

Pengujian dilakukan dengan mengamati pembaruan suhu, mengendalikan LED melalui tombol website dan push button fisik, serta menggeser slider PWM untuk mengetahui perubahan intensitas cahaya LED.

- Serial monitor

```
4.7 | Arduino IDE 3.0.0
File Edit Sketch Tools Help

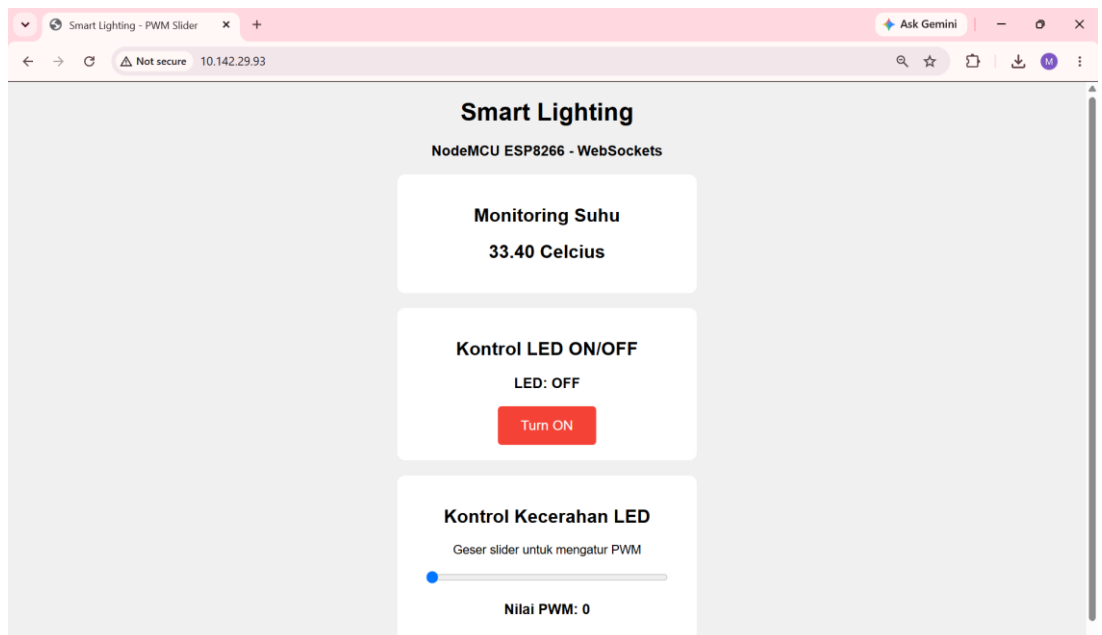
NodeMCU 1.0 (ESP-12...)

4.7.ino
2 #include <ESPAsyncTCP.h>

Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5') New Line 115200 baud

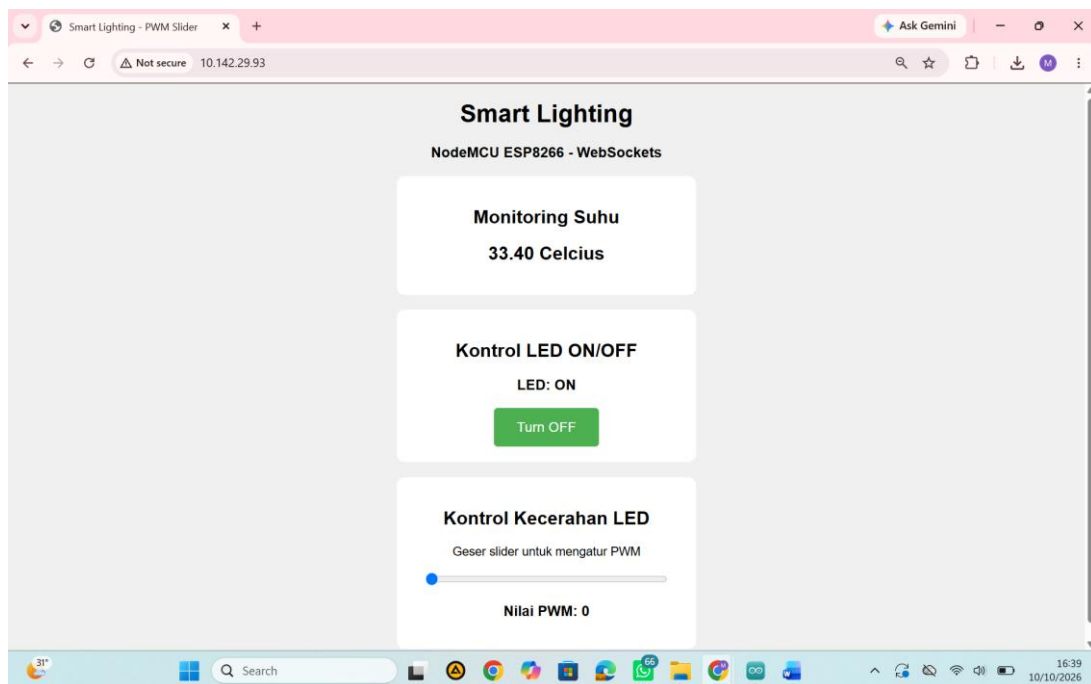
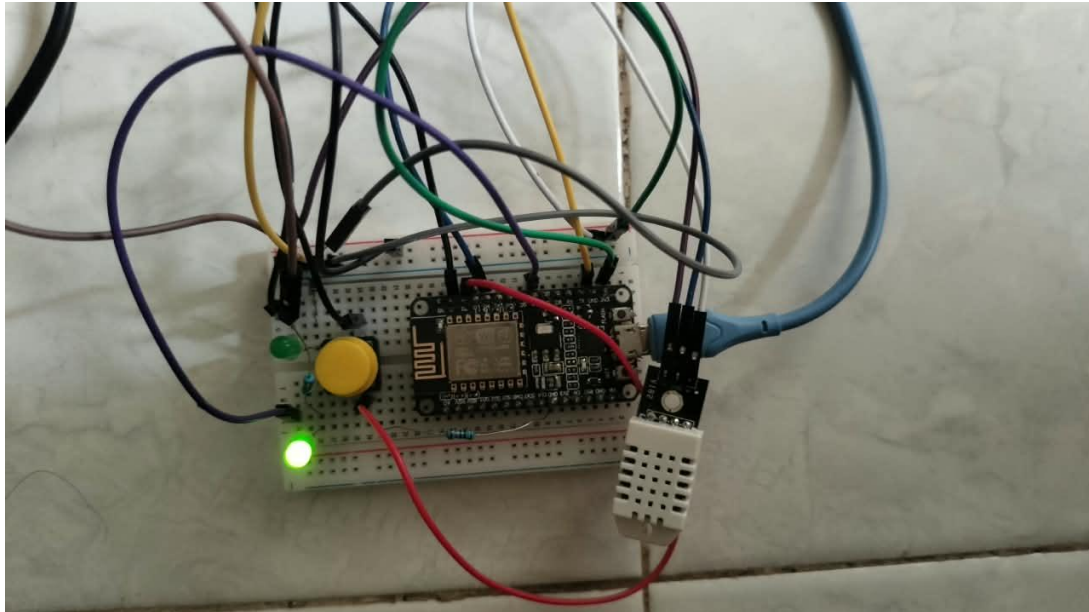
16:46:54.873 -> {{"id": "1", "name": "LED D6 MATI", "pin": 12, "mode": "output"}, {"id": "2", "name": "LED D6 MENYALA", "pin": 13, "mode": "output"}, {"id": "3", "name": "Nilai PWM LED D1:", "pin": 1, "mode": "analog"}]}
16:46:55.587 -> .....
16:47:02.798 -> WiFi Terhubung!
16:47:02.798 -> IP Address: 10.142.29.93
16:47:02.798 -> Web Server Siap!
16:47:02.798 -> Sistem PWM Slider Berjalan!
16:47:30.475 -> Client WebSocket terhubung
16:47:31.728 -> LED D6 MATI
16:47:33.370 -> LED D6 MENYALA
16:47:42.706 -> Nilai PWM LED D1: 172
16:47:43.866 -> Nilai PWM LED D1: 167
16:47:43.900 -> Nilai PWM LED D1: 159
16:47:43.900 -> Nilai PWM LED D1: 149
16:47:43.932 -> Nilai PWM LED D1: 139
16:47:43.932 -> Nilai PWM LED D1: 131
16:47:43.968 -> Nilai PWM LED D1: 121
16:47:44.002 -> Nilai PWM LED D1: 111
16:47:44.002 -> Nilai PWM LED D1: 98
16:47:44.002 -> Nilai PWM LED D1: 88
16:47:44.045 -> Nilai PWM LED D1: 76
16:47:44.077 -> Nilai PWM LED D1: 58
16:47:44.077 -> Nilai PWM LED D1: 45
16:47:44.077 -> Nilai PWM LED D1: 25
16:47:44.115 -> Nilai PWM LED D1: 12
16:47:44.115 -> Nilai PWM LED D1: 0
16:47:44.115 -> Nilai PWM LED D1: 0
```

- Hasil Tampilan Website



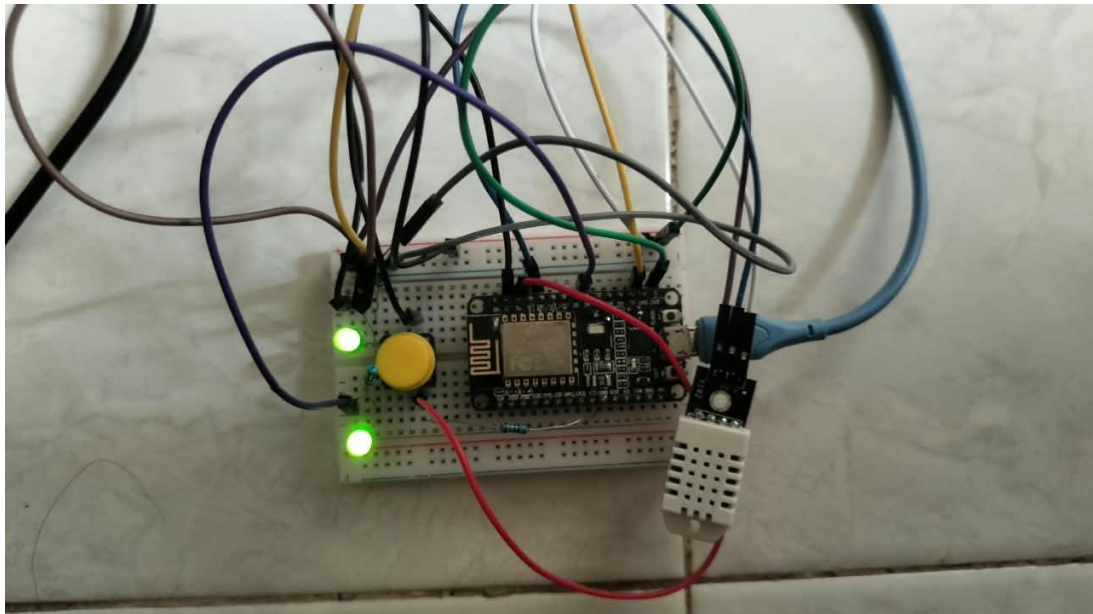
C. Hasil Praktikum

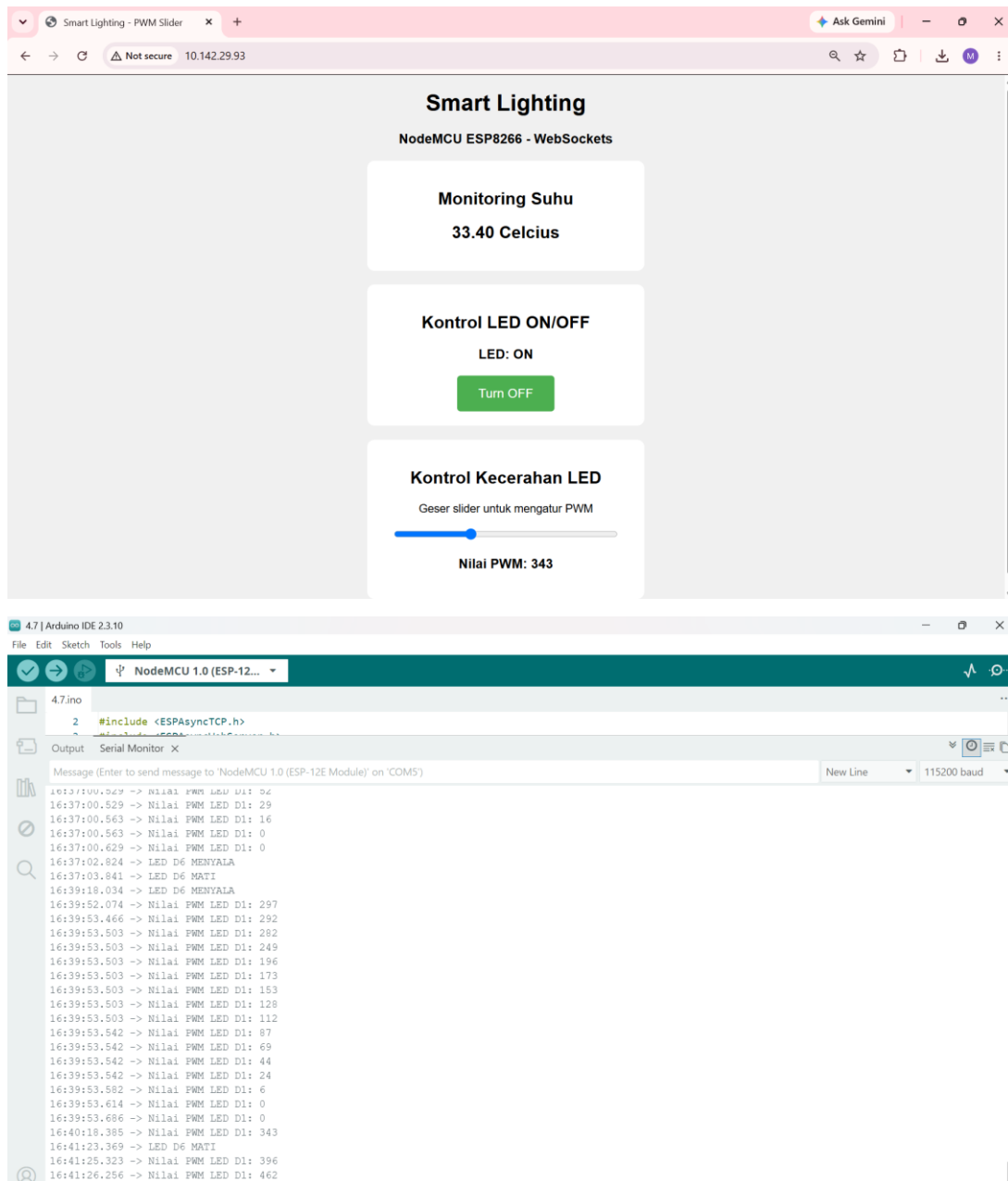
a. Kondisi LED Menyala (ON) ketika di tekan push button ataupun di klik pada bagian web nya TURN ON



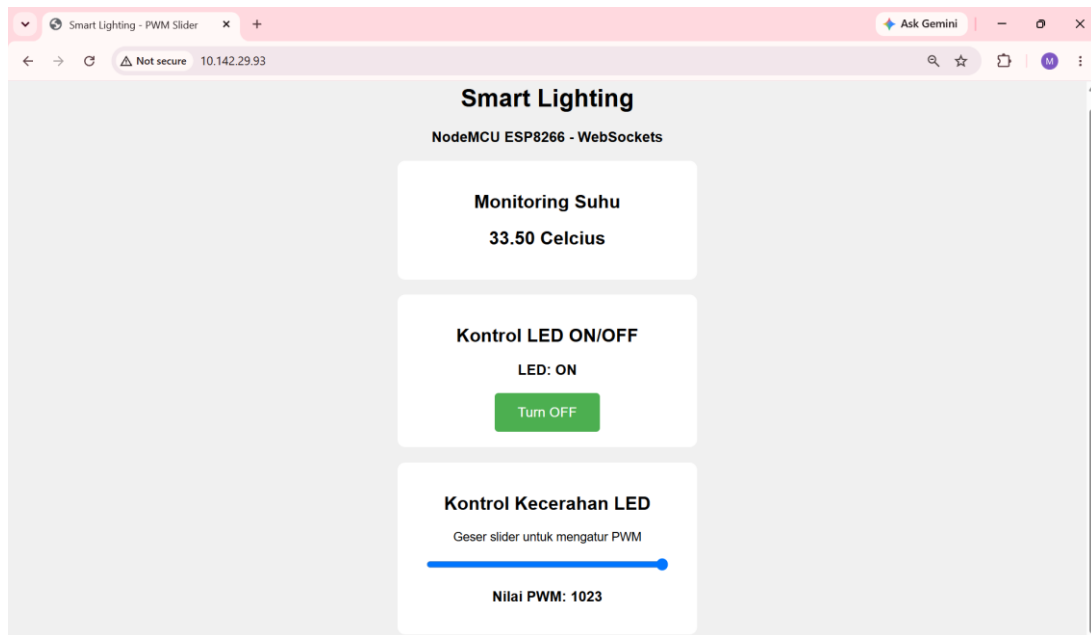
```
16:44:58.230 -> LED D6 MENYALA
16:44:59.881 -> Nilai PWM LED D1: 456
16:44:59.881 -> Nilai PWM LED D1: 446
16:44:59.881 -> Nilai PWM LED D1: 428
16:44:59.881 -> Nilai PWM LED D1: 415
16:44:59.881 -> Nilai PWM LED D1: 390
16:44:59.881 -> Nilai PWM LED D1: 367
16:44:59.920 -> Nilai PWM LED D1: 299
16:44:59.920 -> Nilai PWM LED D1: 248
16:44:59.920 -> Nilai PWM LED D1: 207
16:44:59.953 -> Nilai PWM LED D1: 134
16:44:59.985 -> Nilai PWM LED D1: 73
16:44:59.985 -> Nilai PWM LED D1: 0
16:44:59.985 -> Nilai PWM LED D1: 0
```

b. Kondisi LED Menyala (ON) ketika di tekan push button ataupun di klik pada bagian web nya TURN ON dan juga ketika nilai PWM diatas 0





Dan nilai maksimal yang digunakan di pwm adalah 1023, seperti gambar pada web berikut



Modul 5: Topologi Jaringan Terdesentralisasi (ESP-NOW dan Mesh)

5.7 Tugas: Sistem Telemetry & Otomasi Terdistribusi Berbasis painlessMesh

A. Alat dan Bahan

Alat dan bahan yang digunakan adalah sebagai berikut.

No.	Alat dan Bahan	Jumlah
1	NodeMCU ESP8266	3 buah
2	Sensor DHT11	1 buah
3	Sensor LDR	1 buah
4	Resistor 10k Ohm	1 buah
5	Resistor 220 Ohm	1 buah
6	LED 5mm	1 buah
7	Breadboard	3 buah
8	Kabel jumper	Secukupnya
9	Kabel Micro-USB	3 buah
10	Laptop dengan Arduino IDE	3 unit

B. Langkah Kerja

1. Perancangan Rangkaian

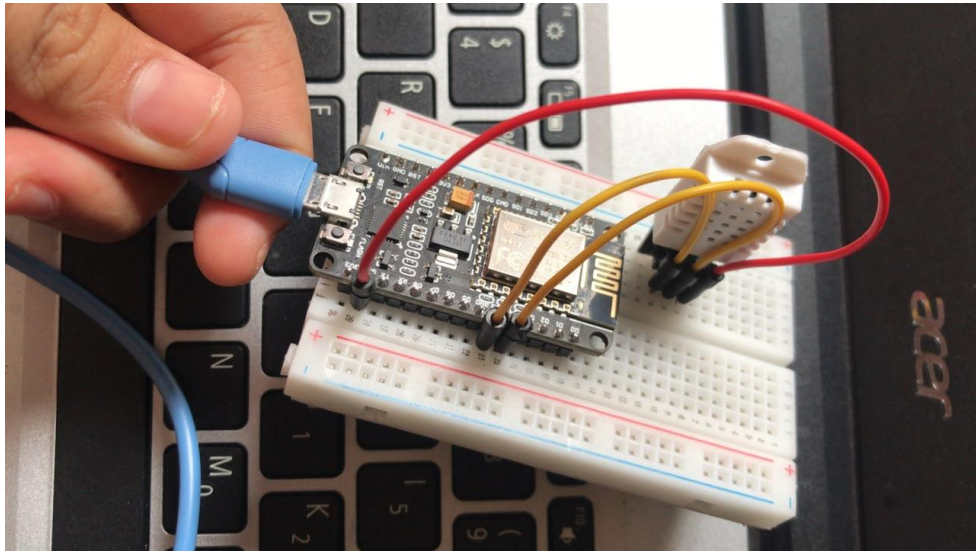
Pada tugas ini dilakukan perancangan sistem telemetry dan otomasi terdistribusi menggunakan tiga perangkat NodeMCU ESP8266 yang saling berkomunikasi melalui jaringan painlessMesh tanpa memerlukan router Wi-Fi eksternal. Pengerjaan dilakukan secara berkelompok dengan pembagian fungsi dan tanggung jawab pada masing-masing node.

Pembagian tugas kelompok adalah sebagai berikut.

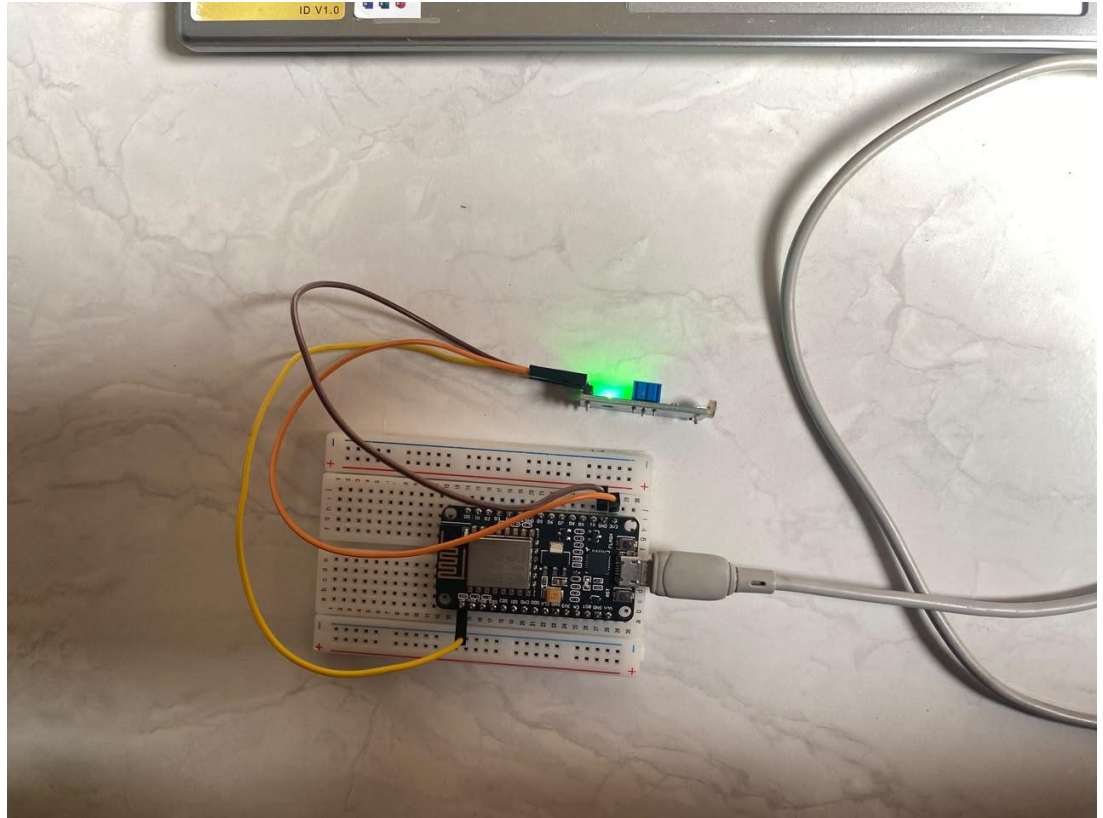
Node	Penanggung Jawab	Fungsi
Node 1	Virna	Membaca suhu dan kelembapan menggunakan sensor DHT11
Node 2	Nayla	Membaca intensitas cahaya menggunakan sensor LDR
Node 3	Sekar	Menerima data sensor dan mengendalikan LED secara otomatis

Pada tahap perancangan, setiap NodeMCU dirangkai sesuai dengan fungsi yang telah ditentukan.

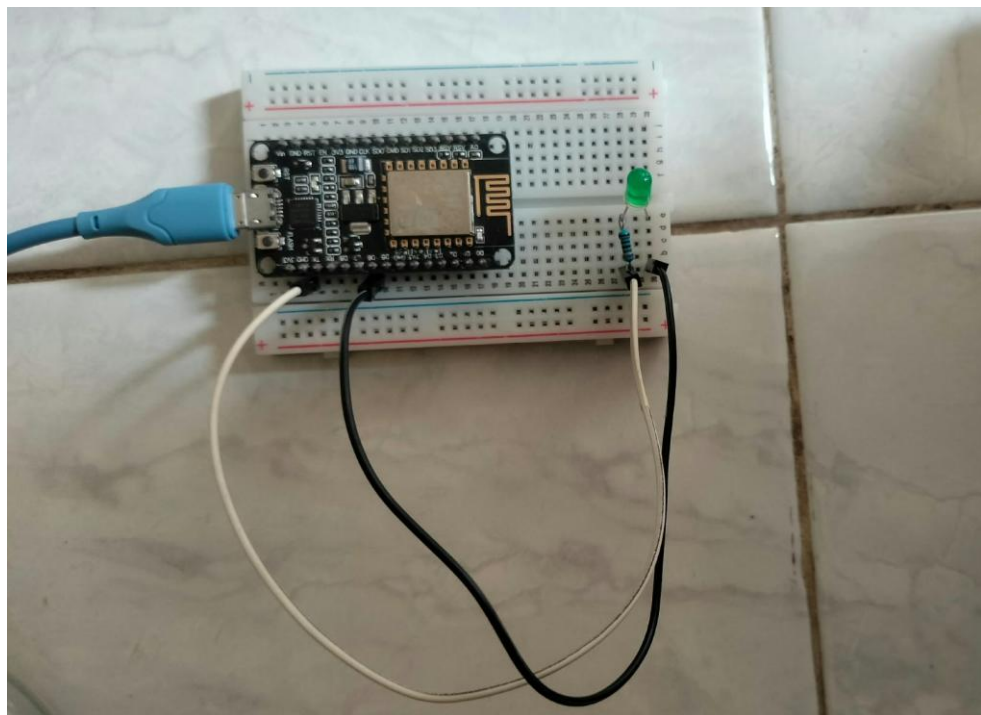
- Node 1 (Sensor DHT11): Sensor DHT11 dihubungkan ke pin D4 (GPIO 2) sebagai jalur data, sedangkan VCC dan GND dihubungkan ke pin 3V3 dan GND NodeMCU. Node ini digunakan untuk membaca suhu dan kelembapan lingkungan.



- Node 2 (Sensor LDR): Sensor LDR dirangkai menggunakan resistor 10k Ohm sebagai pembagi tegangan yang dihubungkan ke pin A0 NodeMCU. Node ini digunakan untuk membaca nilai ADC berdasarkan intensitas cahaya di lingkungan sekitar.



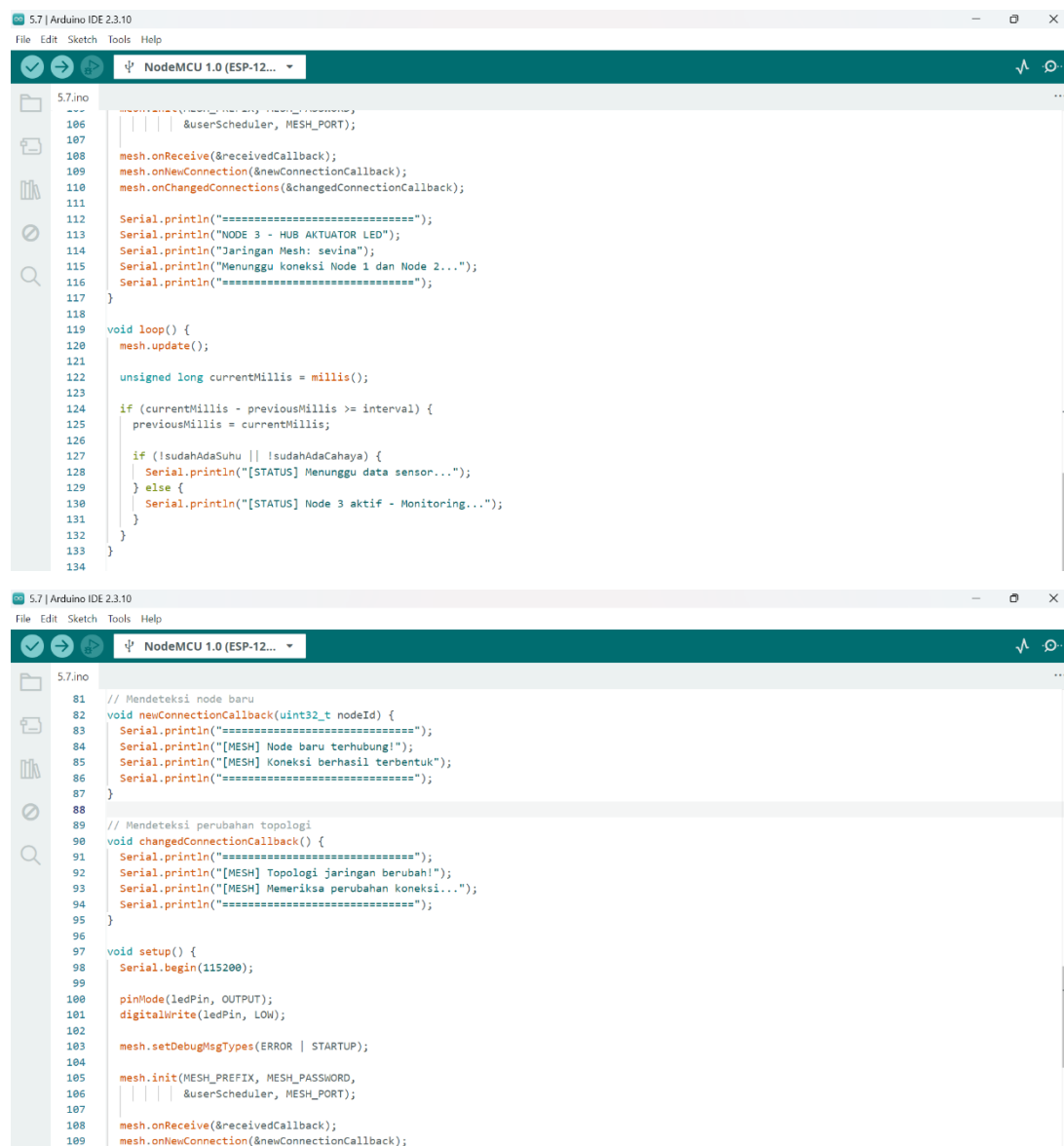
- Node 3 (Hub Aktuator LED): LED dihubungkan ke pin D6 (GPIO 12) NodeMCU dengan resistor 220 Ohm sebagai pembatas arus. Node ini bertugas menerima data dari Node 1 dan Node 2 serta menentukan kondisi LED berdasarkan nilai sensor yang diterima.



Seluruh NodeMCU dikonfigurasi menggunakan nama jaringan, password, dan port yang sama agar dapat membentuk jaringan painlessMesh dan melakukan pertukaran data.

2. Kode yang Digunakan

Pada laporan ini, hanya ditampilkan beberapa bagian penting dari kode program Node 3 yang menjadi tanggung jawab penulis. Hal ini dilakukan karena kode program secara keseluruhan memiliki jumlah baris yang cukup banyak. Adapun kode lengkap dari Node 1, Node 2, dan Node 3 telah diunggah ke repositori GitHub sebagai dokumentasi keseluruhan sistem.



```
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...)

5.7.ino
106 // mesh.onReceive(&receivedCallback);
107 // mesh.onNewConnection(&newConnectionCallback);
108 // mesh.onChangedConnections(&changedConnectionCallback);
109
110 Serial.println("=====");
111 Serial.println("NODE 3 - HUB AKTUATOR LED");
112 Serial.println("Jaringan Mesh: sevina");
113 Serial.println("Menunggu koneksi Node 1 dan Node 2...");
114 Serial.println("=====");
115 }
116
117 void loop() {
118   mesh.update();
119
120   unsigned long currentMillis = millis();
121
122   if (currentMillis - previousMillis >= interval) {
123     previousMillis = currentMillis;
124
125     if (!IsudahAdaSuhu || !IsudahAdaCahaya) {
126       Serial.println("[STATUS] Menunggu data sensor...");
127     } else {
128       Serial.println("[STATUS] Node 3 aktif - Monitoring...");
129     }
130   }
131 }
132
133
134
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...)

5.7.ino
81 // Mendeteksi node baru
82 void newConnectionCallback(uint32_t nodeId) {
83   Serial.println("=====");
84   Serial.println("[MESH] Node baru terhubung!");
85   Serial.println("[MESH] Koneksi berhasil terbentuk");
86   Serial.println("=====");
87 }
88
89 // Mendeteksi perubahan topologi
90 void changedConnectionCallback() {
91   Serial.println("=====");
92   Serial.println("[MESH] Topologi jaringan berubah!");
93   Serial.println("[MESH] Memeriksa perubahan koneksi...");
94   Serial.println("=====");
95 }
96
97 void setup() {
98   Serial.begin(115200);
99
100   pinMode(ledPin, OUTPUT);
101   digitalWrite(ledPin, LOW);
102
103   mesh.setDebugMsgTypes(ERROR | STARTUP);
104
105   mesh.init(MESH_PREFIX, MESH_PASSWORD,
106           &userScheduler, MESH_PORT);
107
108   mesh.onReceive(&receivedCallback);
109   mesh.onNewConnection(&newConnectionCallback);
```

```
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...)

5.7.ino
--
53 if (tipe == "suhu_node") {
54   Serial.println("[TERIMA] Node 1 - Sensor DHT");
55
56   suhuTerakhir = doc["suhu"];
57   float kelembapan = doc["kelembapan"];
58   sudahAdaSuhu = true;
59
60   Serial.printf("Suhu      : %.2f C\n", suhuTerakhir);
61   Serial.printf("Kelembapan : %.2f %%\n", kelembapan);
62 }
63 else if (tipe == "cahaya_node") {
64   Serial.println("[TERIMA] Node 2 - Sensor LDR");
65
66   adcTerakhir = doc["adc"];
67   sudahAdaCahaya = true;
68
69   Serial.printf("ADC Cahaya : %d\n", adcTerakhir);
70 }
71 else {
72   Serial.println("[ERROR] Tipe data tidak dikenal");
73   return;
74 }
75
76 cekKondisiLED();
77
78 Serial.println("=====");
79 }
80 }
```

```
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...)

5.7.ino
28
29 if (suhuPanas || cahayaGelap) {
30   digitalWrite(ledPin, HIGH);
31   Serial.println("LED MENYALA");
32 } else {
33   digitalWrite(ledPin, LOW);
34   Serial.println("LED MATI");
35 }
36 }
37
38 // Menerima dan memproses data JSON
39 void receivedCallback(uint32_t from, String &msg) {
40   StaticJsonDocument<200> doc;
41
42   DeserializationError error = deserializeJson(doc, msg);
43
44   if (error) {
45     Serial.println("[ERROR] Gagal membaca JSON!");
46     return;
47   }
48
49   String tipe = doc["tipe"].as<String>();
50
51   Serial.println("=====");
52
53   if (tipe == "suhu_node") {
54     Serial.println("[TERIMA] Node 1 - Sensor DHT");
55   }
```

C. Hasil Praktikum

1. Hasil Pengujian Komunikasi Data Sensor

```
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help

NodeMCU 1.0 (ESP-12...)

5.7.ino
2 #include <painlessMesh.h>

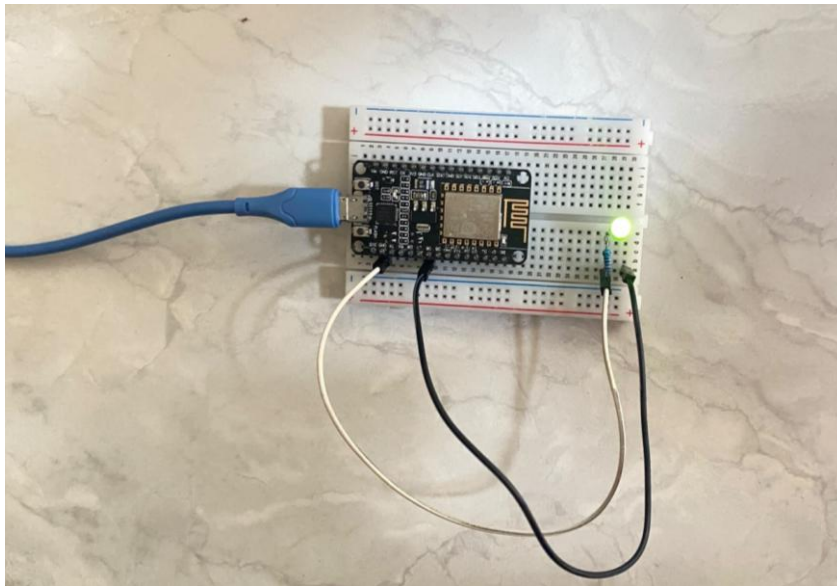
Output Serial Monitor X

Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

15:41:10.375 -> =====
15:41:10.375 -> [TERIMA] Node 2 - Sensor LDR
15:41:10.375 -> ADC Cahaya : 387
15:41:10.375 -> LED MENYALA
15:41:10.375 -> =====
15:41:12.330 -> =====
15:41:12.363 -> [TERIMA] Node 1 - Sensor DHT
15:41:12.363 -> Suhu      : 34.10 C
15:41:12.363 -> Kelembapan : 62.80 %
15:41:12.363 -> LED MENYALA
15:41:12.363 -> =====
15:41:12.592 -> [STATUS] Node 3 aktif - Monitoring...
15:41:15.341 -> =====
```

2. Hasil Pengujian Otomatisasi LED

a. Kondisi LED Menyala



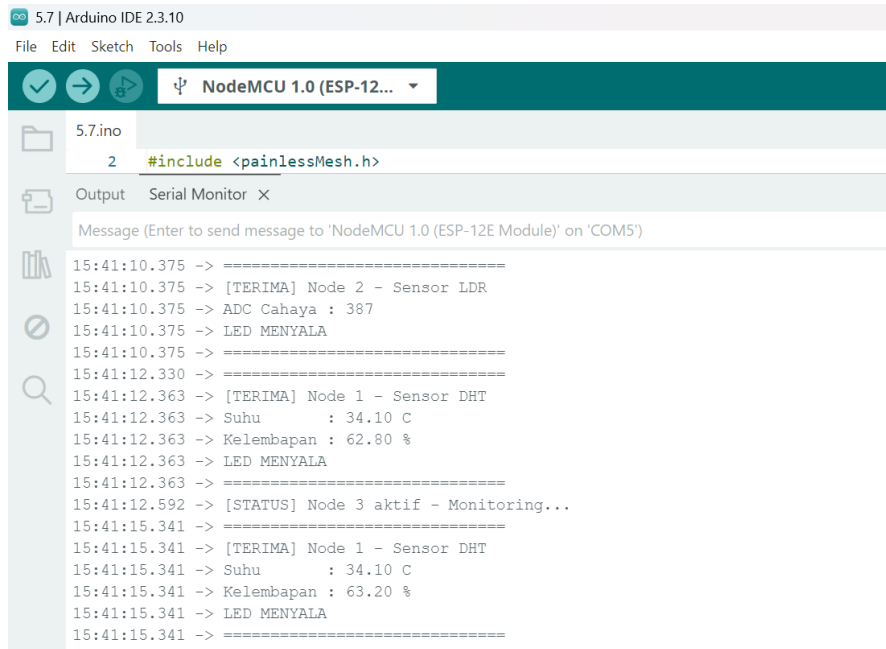
Serial Monitor node 3:

```
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
5.7.ino
2 #include <painlessMesh.h>
Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
15:41:10.375 -> =====
15:41:10.375 -> [TERIMA] Node 2 - Sensor LDR
15:41:10.375 -> ADC Cahaya : 387
15:41:10.375 -> LED MENYALA
15:41:10.375 -> =====
15:41:12.330 -> =====
15:41:12.363 -> [TERIMA] Node 1 - Sensor DHT
15:41:12.363 -> Suhu      : 34.10 C
15:41:12.363 -> Kelembapan : 62.80 %
15:41:12.363 -> LED MENYALA
15:41:12.363 -> =====
15:41:12.592 -> [STATUS] Node 3 aktif - Monitoring...
15:41:15.341 -> =====
15:41:15.341 -> [TERIMA] Node 1 - Sensor DHT
15:41:15.341 -> Suhu      : 34.10 C
15:41:15.341 -> Kelembapan : 63.20 %
15:41:15.341 -> LED MENYALA
15:41:15.341 -> =====
```

3. Hasil Pengujian Multi-Hop Relay dan Self-Healing

a. Kondisi Sebelum Node 2 Diputus

Tampilan node 3

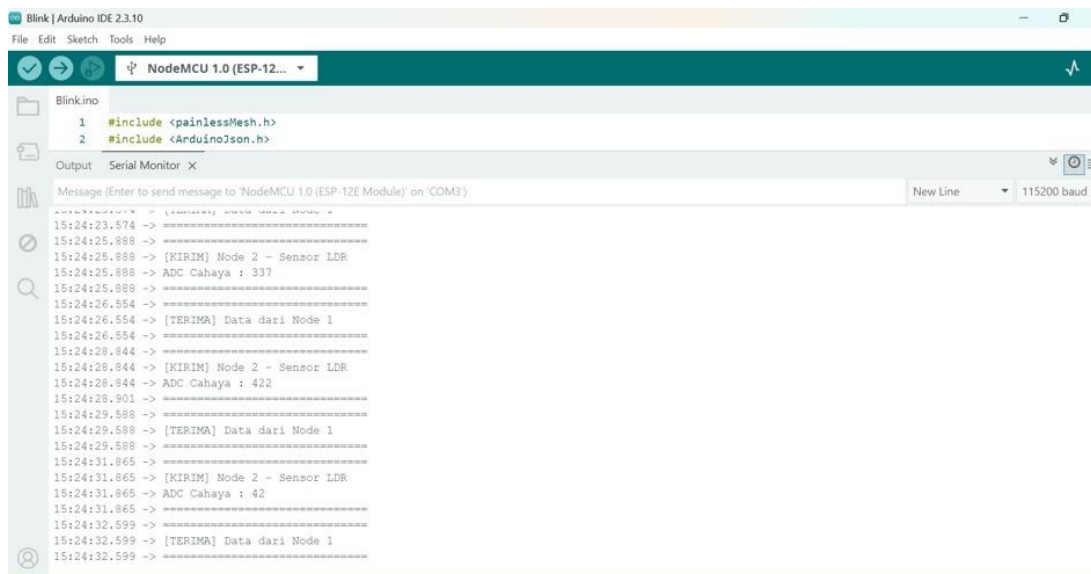


```
5.7.ino
2 #include <painlessMesh.h>

Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')

15:41:10.375 -> =====
15:41:10.375 -> [TERIMA] Node 2 - Sensor LDR
15:41:10.375 -> ADC Cahaya : 387
15:41:10.375 -> LED MENYALA
15:41:10.375 -> =====
15:41:12.330 -> =====
15:41:12.363 -> [TERIMA] Node 1 - Sensor DHT
15:41:12.363 -> Suhu : 34.10 C
15:41:12.363 -> Kelembapan : 62.80 %
15:41:12.363 -> LED MENYALA
15:41:12.363 -> =====
15:41:12.592 -> [STATUS] Node 3 aktif - Monitoring...
15:41:15.341 -> =====
15:41:15.341 -> [TERIMA] Node 1 - Sensor DHT
15:41:15.341 -> Suhu : 34.10 C
15:41:15.341 -> Kelembapan : 63.20 %
15:41:15.341 -> LED MENYALA
15:41:15.341 -> =====
```

Tampilan node 2:



```
Blink.ino
1 #include <painlessMesh.h>
2 #include <ArduinoJson.h>

Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM3') New Line 115200 baud

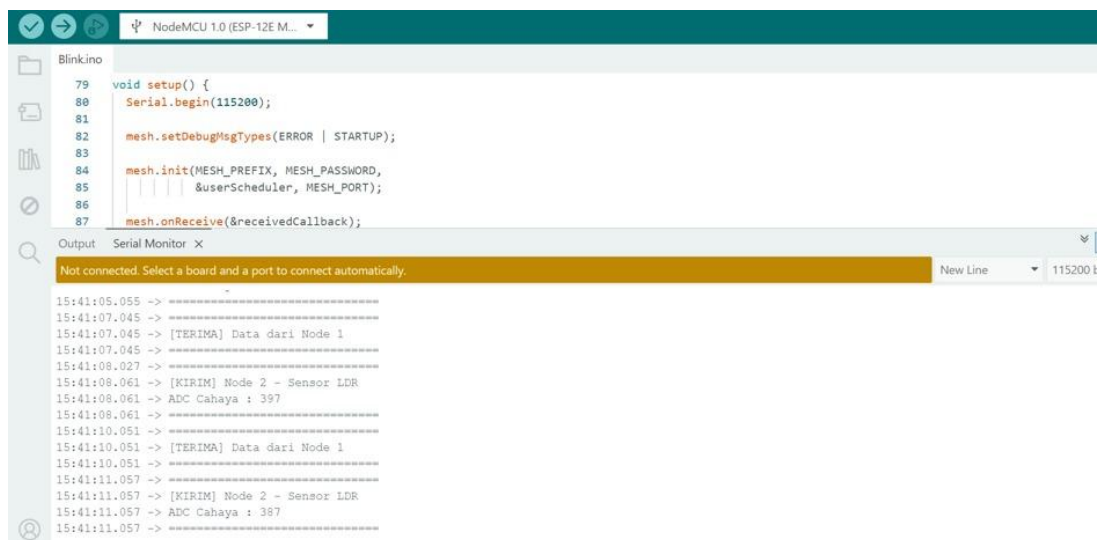
15:24:23.574 -> =====
15:24:25.888 -> =====
15:24:25.888 -> [KIRIM] Node 2 - Sensor LDR
15:24:25.888 -> ADC Cahaya : 337
15:24:25.888 -> =====
15:24:26.554 -> =====
15:24:26.554 -> [TERIMA] Data dari Node 1
15:24:26.554 -> =====
15:24:28.844 -> =====
15:24:28.844 -> [KIRIM] Node 2 - Sensor LDR
15:24:28.844 -> ADC Cahaya : 422
15:24:28.901 -> =====
15:24:29.588 -> =====
15:24:29.588 -> [TERIMA] Data dari Node 1
15:24:29.588 -> =====
15:24:31.865 -> =====
15:24:31.865 -> [KIRIM] Node 2 - Sensor LDR
15:24:31.865 -> ADC Cahaya : 42
15:24:31.865 -> =====
15:24:32.599 -> =====
15:24:32.599 -> [TERIMA] Data dari Node 1
15:24:32.599 -> =====
```

Tampilan node 1:

```
15:30:58.015 -> =====
15:30:58.015 -> [TERIMA] Data dari Node 2
15:30:58.015 -> =====
15:31:00.017 -> =====
15:31:00.017 -> [KIRIM] Node 1 - Sensor DHT
15:31:00.017 -> Suhu      : 34.20 C
15:31:00.017 -> Kelembapan : 63.30 %
15:31:00.017 -> =====
15:31:01.021 -> =====
15:31:01.021 -> [TERIMA] Data dari Node 2
15:31:01.021 -> =====
15:31:02.999 -> =====
15:31:03.032 -> [KIRIM] Node 1 - Sensor DHT
15:31:03.032 -> Suhu      : 34.20 C
15:31:03.032 -> Kelembapan : 63.40 %
15:31:03.032 -> =====
15:31:04.032 -> =====
15:31:04.032 -> [TERIMA] Data dari Node 2
15:31:04.032 -> =====
-- -- -- -- --
```

b. Kondisi Ketika Node 2 Diputus

Tampilan node 2:



```
void setup() {
  Serial.begin(115200);

  mesh.setDebugMsgTypes(ERROR | STARTUP);

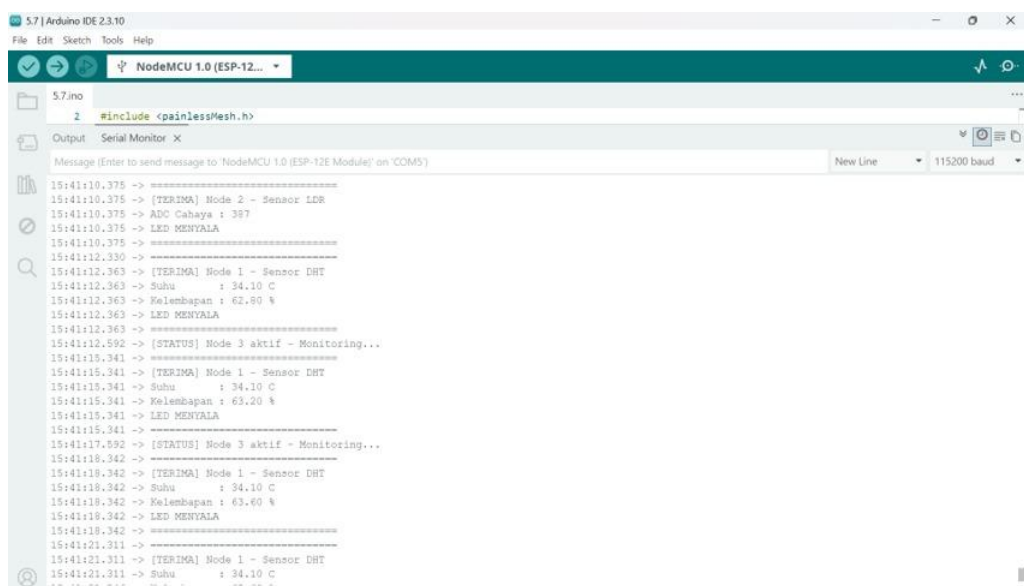
  mesh.init(MESH_PREFIX, MESH_PASSWORD,
            &userScheduler, MESH_PORT);

  mesh.onReceive(&receivedCallback);
}
```

Not connected. Select a board and a port to connect automatically.

```
15:41:05.055 -> =====
15:41:07.045 -> =====
15:41:07.045 -> [TERIMA] Data dari Node 1
15:41:07.045 -> =====
15:41:08.027 -> =====
15:41:08.061 -> [KIRIM] Node 2 - Sensor LDR
15:41:08.061 -> ADC Cahaya : 397
15:41:08.061 -> =====
15:41:10.051 -> =====
15:41:10.051 -> [TERIMA] Data dari Node 1
15:41:10.051 -> =====
15:41:11.057 -> =====
15:41:11.057 -> [KIRIM] Node 2 - Sensor LDR
15:41:11.057 -> ADC Cahaya : 387
15:41:11.057 -> =====
```

Tampilan node 3:



```
#include <painlessMesh.h>

mesh.init(MESH_PREFIX, MESH_PASSWORD,
          &userScheduler, MESH_PORT);

mesh.onReceive(&receivedCallback);
```

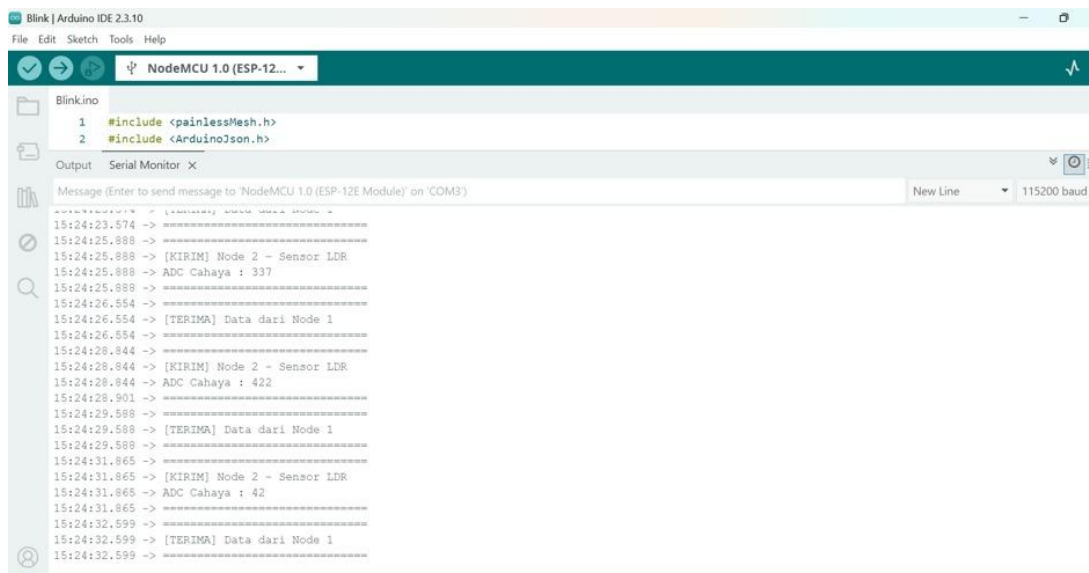
```
15:41:10.375 -> =====
15:41:10.375 -> [TERIMA] Node 2 - Sensor LDR
15:41:10.375 -> ADC Cahaya : 387
15:41:10.375 -> LED MENYALA
15:41:10.375 -> =====
15:41:12.330 -> =====
15:41:12.363 -> [TERIMA] Node 1 - Sensor DHT
15:41:12.363 -> Suhu      : 34.10 C
15:41:12.363 -> Kelembapan : 62.80 %
15:41:12.363 -> LED MENYALA
15:41:12.363 -> =====
15:41:12.592 -> [STATUS] Node 3 aktif - Monitoring...
15:41:15.341 -> =====
15:41:15.341 -> [TERIMA] Node 1 - Sensor DHT
15:41:15.341 -> Suhu      : 34.10 C
15:41:15.341 -> Kelembapan : 63.20 %
15:41:15.341 -> LED MENYALA
15:41:15.341 -> =====
15:41:17.592 -> [STATUS] Node 3 aktif - Monitoring...
15:41:18.342 -> =====
15:41:18.342 -> [TERIMA] Node 1 - Sensor DHT
15:41:18.342 -> Suhu      : 34.10 C
15:41:18.342 -> Kelembapan : 63.60 %
15:41:18.342 -> LED MENYALA
15:41:18.342 -> =====
15:41:21.311 -> =====
15:41:21.311 -> [TERIMA] Node 1 - Sensor DHT
15:41:21.311 -> Suhu      : 34.10 C
15:41:21.311 -> =====
```

Tampilan node 1:

```
15:50:25.837 -> =====
15:50:25.837 -> [KIRIM] Node 1 - Sensor DHT
15:50:25.837 -> Suhu      : 34.00 C
15:50:25.872 -> Kelembapan : 63.40 %
15:50:25.873 -> =====
15:50:28.868 -> =====
15:50:28.868 -> [KIRIM] Node 1 - Sensor DHT
15:50:28.868 -> Suhu      : 34.00 C
15:50:28.868 -> Kelembapan : 63.50 %
15:50:28.868 -> =====
15:50:31.866 -> =====
15:50:31.866 -> [KIRIM] Node 1 - Sensor DHT
15:50:31.866 -> Suhu      : 34.00 C
15:50:31.866 -> Kelembapan : 63.60 %
15:50:31.866 -> =====
15:50:34.854 -> =====
15:50:34.854 -> [KIRIM] Node 1 - Sensor DHT
15:50:34.854 -> Suhu      : 34.00 C
15:50:34.854 -> Kelembapan : 63.60 %
15:50:34.854 -> =====
15:50:37.866 -> =====
15:50:37.866 -> [KIRIM] Node 1 - Sensor DHT
15:50:37.866 -> Suhu      : 34.10 C
15:50:37.866 -> Kelembapan : 63.40 %
```

c. Kondisi Setelah Node 2 Dihubungkan Kembali

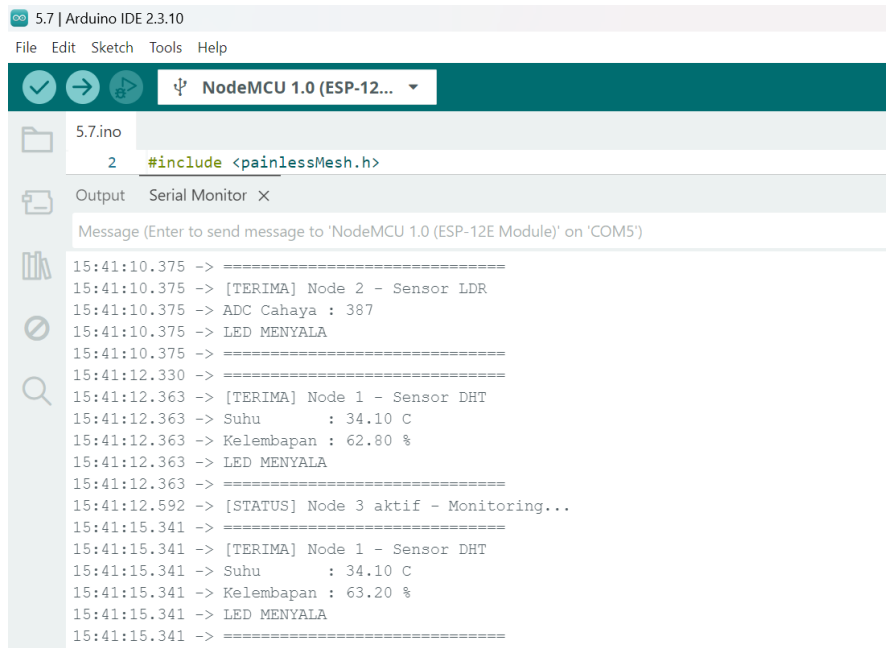
Tampilan node 2:



Tampilan node 3:

```
15:30:58.015 -> =====
15:30:58.015 -> [TERIMA] Data dari Node 2
15:30:58.015 -> =====
15:31:00.017 -> =====
15:31:00.017 -> [KIRIM] Node 1 - Sensor DHT
15:31:00.017 -> Suhu      : 34.20 C
15:31:00.017 -> Kelembapan : 63.30 %
15:31:00.017 -> =====
15:31:01.021 -> =====
15:31:01.021 -> [TERIMA] Data dari Node 2
15:31:01.021 -> =====
15:31:02.999 -> =====
15:31:03.032 -> [KIRIM] Node 1 - Sensor DHT
15:31:03.032 -> Suhu      : 34.20 C
15:31:03.032 -> Kelembapan : 63.40 %
15:31:03.032 -> =====
15:31:04.032 -> =====
15:31:04.032 -> [TERIMA] Data dari Node 2
15:31:04.032 -> =====
-- -- -- --
```

Tampilan node 1:



```
5.7 | Arduino IDE 2.3.10
File Edit Sketch Tools Help
NodeMCU 1.0 (ESP-12...
5.7.ino
2 #include <painlessMesh.h>
Output Serial Monitor X
Message (Enter to send message to 'NodeMCU 1.0 (ESP-12E Module)' on 'COM5')
15:41:10.375 -> =====
15:41:10.375 -> [TERIMA] Node 2 - Sensor LDR
15:41:10.375 -> ADC Cahaya : 387
15:41:10.375 -> LED MENYALA
15:41:10.375 -> =====
15:41:12.330 -> =====
15:41:12.363 -> [TERIMA] Node 1 - Sensor DHT
15:41:12.363 -> Suhu : 34.10 C
15:41:12.363 -> Kelembapan : 62.80 %
15:41:12.363 -> LED MENYALA
15:41:12.363 -> =====
15:41:12.592 -> [STATUS] Node 3 aktif - Monitoring...
15:41:15.341 -> =====
15:41:15.341 -> [TERIMA] Node 1 - Sensor DHT
15:41:15.341 -> Suhu : 34.10 C
15:41:15.341 -> Kelembapan : 63.20 %
15:41:15.341 -> LED MENYALA
15:41:15.341 -> =====
```

Hasil pengamatan waktu selama pengujian dicatat pada tabel berikut.

No.	Kondisi Pengujian	Waktu Pengamatan
1	Ketiga node terhubung	15:30:00
2	Node 2 diputus	15:30:08
3	Perubahan topologi terdeteksi	15:30:17
4	Node 2 dinyalakan kembali	15:30:30
5	Node 2 kembali terhubung	15:30:35
6	Data sensor kembali diterima	15:30:45

Berdasarkan hasil pengujian, jaringan painlessMesh membutuhkan waktu 9 detik untuk mendeteksi perubahan topologi setelah Node 2 diputus dari sumber daya. Ketika Node 2 dinyalakan kembali, perangkat membutuhkan waktu 5 detik untuk terhubung kembali ke jaringan. Selanjutnya, data sensor kembali diterima oleh Node 3 pada pukul 15:30:45, yaitu 15 detik setelah Node 2 dinyalakan kembali. Hasil pengamatan ini menunjukkan bahwa jaringan painlessMesh mampu membentuk kembali koneksi secara otomatis setelah terjadi pemutusan node, sehingga komunikasi antarnode dapat kembali berlangsung tanpa konfigurasi ulang secara manual.